

Pure ℓ_1 -SVM under Independent Random Label Flips

FA/Banach run candidate partial result

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Status. This is a candidate partial result, not a full answer to every noise model in the source question. It proves directional consistency for the original pure ℓ_1 constraint under independent symmetric label flips.

1 Source question

Kolleck and Vybřál study hinge-risk minimization over an ℓ_1 ball [1]. In Section VI (Discussion), page 15 of the arXiv PDF, they ask for the behavior of ℓ_1 -SVM when a small probability of label misclassification is present. They also ask whether their dependence of the sample count on the accuracy and Gaussian scale is optimal.

errors between a and $\hat{a}/\|\hat{a}\|_2$ for $n = 60$ trials.

We indeed see that to achieve the same error, the number of measurements only needs to grow logarithmically in d , explaining once again the success of ℓ_1 -SVM for high-dimensional classification problems.

VI. DISCUSSION

In this paper we have analyzed the performance of ℓ_1 -SVM (II.5) in recovering sparse classifiers. Theorem II.3 shows, that a good approximation of such a sparse classifier can be achieved with small number of learning points m if the data is well spread. The geometric properties of well distributed learning points are modelled by independent Gaussian vectors with growing variance r and it would be interesting to know, how ℓ_1 -SVM performs on points chosen independently from other distributions. The number of learning points needs to grow logarithmically with the underlying dimension d and linearly with the sparsity of the classifier. On the other hand, the optimality of the dependence of m on ε and r remains open. Another important question left open is the behavior of ℓ_1 -SVM in the presence of missclassifications, i.e. when there is a (small) probability that the signs $y_i \in \{-1, +1\}$ do not coincide with $\text{sign}(\langle x_i, a \rangle)$. Finally, we proposed a modification of ℓ_1 -SVM by incorporating an additional ℓ_2 -constraint.

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REFERENCES

- [1] A. Ai, A. Lapanowski, Y. Plan, and R. Vershynin, "One-bit compressed sensing with non-Gaussian measurements", *Linear Algebra Appl.*, vol. 441, pp. 222–239, 2014.
- [2] K.P. Bennett and O.L. Mangasarian, "Robust linear programming discrimination of two linearly independent inseparable sets", *Optimization Methods and Software* on 23–34 1992

The result below treats the independent random-flip subcase of the first question. It keeps the source estimator unchanged: there is no added ℓ_2 constraint.

2 Model and theorem

Fix $d \geq 2$, $r, R > 0$, and $a \in \mathbb{R}^d$ with $\|a\|_2 = 1$ and $\|a\|_1 \leq R$. Let g_i be independent standard Gaussian vectors, set $x_i = rg_i$, and let $\xi_i \in \{-1, 1\}$ be independent of the g_i with

$$\mathbb{P}(\xi_i = 1) = p \in (1/2, 1).$$

The observed labels and empirical risk are

$$y_i = \xi_i \operatorname{sign}\langle g_i, a \rangle, \quad L_m(w) = \frac{1}{m} \sum_{i=1}^m [1 - y_i \langle x_i, w \rangle]_+.$$

The ordinary ℓ_1 -SVM minimizes L_m over $K = \{w : \|w\|_1 \leq R\}$.

Put

$$c_p = \sqrt{2 \log \frac{p}{2p-1}}, \quad \mu = \frac{1}{rc_p},$$

and assume the feasibility condition $\mu\|a\|_1 \leq R$. Define

$$\phi_p(t) = p[1 - t]_+ + (1 - p)[1 + t]_+.$$

If $X = |G|$ and Z are independent, with $G, Z \sim N(0, 1)$, set

$$H(u, s) = \mathbb{E} \phi_p(r(uX + sZ)), \quad u \in \mathbb{R}, \quad s \geq 0.$$

For $0 < \delta < \mu/2$, define the explicit separation modulus

$$\Delta_\delta = \min\{H(u, s) - H(\mu, 0) : |u| \leq R, \ 0 \leq s \leq R, \\ (u - \mu)^2 + s^2 \geq \delta^2\}.$$

Theorem 1 (Random-flip consistency for pure ℓ_1 -SVM). *The population risk $L(w) = \mathbb{E}L_m(w)$ has the unique global minimizer $w_\star = \mu a$ on all of $\mathbb{R}^d$. In particular, $w_\star$ is the unique minimizer on K , and $\Delta_\delta > 0$.*

Let

$$A_{d,r,R} = 8\sqrt{8\pi} + 18rR\sqrt{2\log(2d)}.$$

If

$$m \geq \frac{64A_{d,r,R}^2}{\Delta_\delta^2},$$

then, with probability at least

$$1 - 8 \left(e^{-m\Delta_\delta^2/2048} + e^{-m\Delta_\delta^2/(2048r^2R^2)} \right),$$

every empirical minimizer $\hat{a} \in \arg \min_K L_m$ satisfies

$$\|\hat{a} - \mu a\|_2 < \delta, \quad \left\| \frac{\hat{a}}{\|\hat{a}\|_2} - a \right\|_2 < \frac{2\delta}{\mu}.$$

3 Proof intuition

Write a candidate as $w = ua + v$ with $v \perp a$. Gaussian rotation invariance reduces the signed margin to $r(u|G| + \|v\|_2 Z)$. The noisy hinge profile ϕ_p is convex. Adding the independent centered Gaussian term caused by v therefore raises population risk, strictly if $v \neq 0$. The problem collapses to a one-dimensional profile $J(u) = H(u, 0)$, whose derivative has a single zero at the displayed value μ . Uniform empirical concentration, already proved for the source ℓ_1 -SVM, then lifts the unique population minimizer to a finite-sample directional statement.

4 Proof

Decompose $w = ua + v$, where $u = \langle w, a \rangle$ and $v \perp a$, and put $s = \|v\|_2$. Write a standard Gaussian g as

$$g = G_0 a + g_\perp.$$

The random variables $|G_0|$ and $\text{sign}(G_0)\langle g_\perp, v/s \rangle$ are independent; the second is standard Gaussian when $s > 0$. Consequently,

$$\text{sign}\langle g, a \rangle \langle rg, w \rangle \stackrel{d}{=} r(uX + sZ),$$

and hence $L(w) = H(u, s)$.

The function ϕ_p is convex and is not affine on $\mathbb{R}$: its slopes change at -1 and 1 . Conditional on X , Jensen's inequality gives

$$H(u, s) \geq H(u, 0) =: J(u).$$

If $s > 0$, the conditional Gaussian variable $r(uX + sZ)$ has full support, so equality in Jensen is impossible. Thus the inequality is strict for $s > 0$.

It remains to minimize J . For $u < 0$, the argument ruX is nonpositive and all slopes of ϕ_p there are negative because $p > 1/2$. Hence J is strictly decreasing on $(-\infty, 0]$. For $u > 0$, differentiation under the expectation is legitimate because X has no atoms at the kink, and

$$\begin{aligned} J'(u) &= r\mathbb{E}X \left((1-p) - p\mathbf{1}_{\{X < 1/(ru)\}} \right) \\ &= r\sqrt{\frac{2}{\pi}} \left(1 - 2p + p \exp\left(-\frac{1}{2r^2u^2}\right) \right). \end{aligned}$$

This expression is strictly increasing in u , has a negative limit at $0+$ and a positive limit at infinity, and vanishes precisely when

$$\exp\left(-\frac{1}{2r^2u^2}\right) = \frac{2p-1}{p},$$

that is, precisely at $u = \mu$. Therefore J has the unique global minimizer μ . Combining this with the strict Jensen step shows that H , and hence L , has the unique global minimizer $(u, s) = (\mu, 0)$, corresponding to $w_\star = \mu a$. The feasibility hypothesis places $w_\star$ in K .

Since $\|w\|_2 \leq \|w\|_1 \leq R$ for $w \in K$, every such w has $|u| \leq R$ and $0 \leq s \leq R$. The parameter set defining Δ_δ is compact, nonempty, and excludes the unique minimizer. Continuity of H therefore implies $\Delta_\delta > 0$.

We now use the source's uniform concentration theorem. Its proof remains valid under the extra independent flips: in the symmetrized process, every factor $\varepsilon_i y_i$ is again an independent Rademacher sign conditional on the data, so the Gaussian and dual- ℓ_1 estimates are unchanged. Thus for every $q > 0$,

$$\sup_{w \in K} |L_m(w) - L(w)| \leq \frac{A_{d,r,R}}{\sqrt{m}} + q$$

with probability at least

$$1 - 8 \left(e^{-mq^2/32} + e^{-mq^2/(32r^2R^2)} \right).$$

Choose $q = \Delta_\delta/8$. The sample hypothesis makes the right-hand side of the uniform bound at most $\Delta_\delta/4$. If $\hat{a}$ minimizes L_m on K , empirical optimality gives

$$L(\hat{a}) - L(w_\star) \leq 2 \sup_{w \in K} |L_m(w) - L(w)| \leq \frac{\Delta_\delta}{2}.$$

By the definition of Δ_δ , this forces $\|\hat{a} - w_\star\|_2 < \delta$. Since $\delta < \mu/2$, the vector $\hat{a}$ is nonzero, and the standard normalization inequality yields

$$\left\| \frac{\hat{a}}{\|\hat{a}\|_2} - a \right\|_2 \leq \frac{2\|\hat{a} - \mu a\|_2}{\mu} < \frac{2\delta}{\mu}.$$

The probability formula follows by substituting $q = \Delta_\delta/8$ into the preceding concentration estimate. $\square$

5 Relation to later literature and limitations

Genzel and Stollenwerk [2] prove sharper noisy hinge-loss recovery estimates with the optimal $m^{-1/2}$ oversampling rate when the convex signal set is contained in the Euclidean unit ball. Their Example 3.4(2) and Proposition 3.5 cover the same independent random-flip law and compute the correlation parameter $(2p-1)\sqrt{2/\pi}$. They explicitly identify Kalleck–Vybřal as a special case and improvement.

The supporting paper also says that extending the noisy argument to arbitrary convex signal sets is not straightforward. The theorem here covers the pure RB_1^d constraint even when it is not contained in B_2^d , but its rate is implicit through Δ_δ . It does not cover adversarial, feature-dependent, or asymmetric flips, non-Gaussian measurements, or prove optimal dependence on p, r , and the target accuracy.

6 Novelty check

The run indexes contained no row for arXiv:1509.08083. Bounded primary-source searches for combinations of random bit flips, ℓ_1 -SVM, Gaussian hinge loss, and symmetric label noise found [2] and the broader symmetric-label-noise literature, but no exact theorem for a pure RB_1^d constraint with the scale formula above. Novelty is plausible and should be checked by a human against papers citing [2].

References

- [1] A. Kalleck and J. Vybřal, *Non-asymptotic Analysis of ℓ_1 -norm Support Vector Machines*, arXiv:1509.08083.
- [2] M. Genzel and A. Stollenwerk, *Robust 1-Bit Compressed Sensing via Hinge Loss Minimization*, arXiv:1804.04846; Information and Inference 9 (2020), 361–422.